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Abstract

The bachelor thesis examines the change in positive framing in sociological publications against the background of academic capitalism. Based on the theory of increasing self-promotion, it is analyzed whether this is reflected in scientific abstracts through increasingly positive language. The paper conducts a sentiment analysis of 121,789 English-language abstracts of sociological articles from the Web of Science database (1993-2023) with two dictionaries for additional validity. Interval plots, linear regressions and Monte Carlo simulations are used to examine temporal trends in positive and negative sentiment. The results show a very small, statistically insignificant increase in positive sentiment over time, while no clear trend can be identified for negative sentiment. The work reveals methodological challenges in the sentiment analysis of scientific texts, particularly with regard to data distribution and the significance interpretation of large samples. Contrary to the results of previous research, the study provides no evidence for a substantial increase in positive framing.

1 Acknowledgements

1.1 Generative AI Statement

This thesis was written with assistance from generative AI tools for specific purposes including literature research, text editing, and formatting. All substantive content, analysis, and conclusions are the result of my own intellectual work. AI-generated suggestions were critically evaluated and modified as appropriate. Complete transparency regarding AI usage has been maintained throughout the research process.

1.2 Ethics Statement

This research was conducted in accordance with the ethical guidelines of Justus Liebig University Giessen and the German Sociological Association. All data used in this study consists of publicly available academic publications. No human subjects were involved in this research, and therefore no ethics approval was required. The analysis was conducted with integrity, transparency, and respect for intellectual property rights.

2 Introduction

2.1 Motivation & Relevance Statement [280w, 0.5p]

- https://publications.jrc.ec.europa.eu/repository/handle/JRC144603
  - decentralization of information is already happening

2.2 Research Question [280w,0.5p]

2.3 Summary of other Sections (Basically summary of Introduction Paragraphs) [550w - 1 page]

Question: How does the usage of populist rhetoric by german Members of Parliament on Twitter affect the politicians engagement communities?

H1: German MPs who use populist rhetoric have more strongly interconnected alters in their reply ego networks than those who do not.

- linguistic construction of in/outgroup
- hyping the ingroup –> selfselection effect keeps only engaged users with knowledge of groups social facts that actively reply to each other
- In results part:
  - In exploratory analysis, we operationalized alter disconnectedness as the fragmentation ratio (number of components divided by number of alters). Results mirror the main finding: lower fragmentation — i.e., more cohesive alter networks — is associated with populist rhetoric (OR = 0.02, p = .017).
  - Exploratory analysis using the fragmentation ratio (components / alters) corroborates this pattern. Populist MPs' reply networks are structurally more cohesive — their alters cluster into fewer, larger communities rather than responding in isolation (OR = 0.02, p = .017). This is consistent with the theoretical expectation that populist rhetoric fosters a sense of shared group identity among respondents, encouraging mutual engagement beyond dyadic replies to the politician.

H2: Do Populist MPs share more audience with each other than with non-populist MPs?

H3: What Network Structures are indicative of Populist MPs ego networks?

3 Literature Review

3.1 Introduction [100w, 0.2page]

3.2 Theoretical Background (start broad) [650w, 1.2 pages]

3.3 Relevant Theory (less broad) [680w, 1.3 pages]

3.4 Case Study (detailed) [670w, 1.3 pages]

3.5 Hypothesis (narrow down) [500w, 1page]

3.6 Summary [100w, 0.2 page]

4 Data and Methods

4.1 Introduction [100w, 0.2p]

4.2 Dataset Sampling & Description [264w, 0.5p]

The dataset comprises tweets and replies directed at German members of parliament, collected over a two-week period from February 7 to February 14, 2022. Data were obtained via Twitter's streaming API using two parallel strategies. The follow stream tracked replies to MP-generated tweets, while the track stream captured tweets mentioning MP handles (e.g., "@username"), including direct and nested replies. Protected accounts were excluded. Both datasets were combined and deduplicated. From the raw data, reply-threads were constructed by linking tweets through directed reply chains. These interactions were then aggregated to a user-level reply network, weighted by the number of exchanges. Thus, the dataset enables an analysis of interaction patterns between politicians and their engagement communities on social media. To deliniate the set of politicians and add their name and party information, a list of parliamentary members of the 19th, 20th, and 21st German Bundestag is obtained from the Bundestags Webarchiv and added to the original dataset through their twitter-account links (Bundestag 2026).

The reply threads are constructed by treating tweets as nodes that are linked through directed replies. A component in the built network is then a replythread with one rootnode that is only recieving links. Some reply chains trace back to MP tweets posted before the collection period, so the root node has no row in the dataset. Matching the target tweets user information against the politician table recovers their authorship, recovering 2,631 of 89,561 threads (2.9%). Threads root tweet information like politician with e.g. party and followerinformation, are then stored within the reply tweets of the respective thread.

4.3 Operationalization & Measure [500w. 1p]

Current studies almost uniformly base their understanding of populism on Mudde (2004)'s definition of populism as [two homogeneous and antagonistic groups, 'the pure people' versus 'the corrupt elite', and which argues that politics should be an expression of the volonté générale (general will) of the people.]. While all operationalizations include a "pro-people" and "anti-elite" one of multiple third dimensions is often implemented aswell like the inclusion of anti-pluralist attitudes, the distinction between leftwing and rightwing populism or agitating against horizontal outgroups like minorities (Aalberg et al. 2017; Castanho Silva et al. 2020; Meyer et al. 2025). Populist Attitudes are not only measured through surveys (Castanho Silva et al. 2020),

but also through observational studies of political discourse on social media (Meyer et al. 2025) leveraging LLMs. This study operationalizes the core dimensions of Mudde (2004)'s definition, People Attitude, Elitist Attitude, and Antagonism, by instructing a large language model through an annotation prompt (see Appendix). Each dimension is defined with explicit scoring anchors: People Attitude and Elitist Attitude are measured on bidirectional scales from –3 to +3, where positive values indicate support for and negative values indicate opposition to the respective group, while Antagonism is measured on a unidirectional scale from 0 (no divide) to 6 (existential threat), with labeled thresholds distinguishing dissatisfaction (1–2), active blame (3–4), and existential threat framing (5–6). The prompt leverages chain-of-thought style reasoning and few-shot examples to guide the models annotation behaviour. To guide the models reasoning letting it question itself throughout the process lead improved the results immensly compared to hard rule-based checks.

The prompt starts with a role definition and a pre-analysis check if the text carries any content other than just a link or a user mention. It proceeds by defining "People Attitude", "Elite Attitude" and "Antagonism". The definition of "the people" is restricted to a broad ordinary majority and explicitly excludes named individuals, lists of specific persons, and narrow subgroups unless the text frames them as standing in for the general public. Similarly, elite criticism is only scored when the target is a generalized powerful class rather than a single individual or a specific policy disagreement. In the last major section the prompt invokes the chain-of-thought before assigning scores, the model must produce a holistic redescription of the post's rhetorical strategy, an actor-by-actor analysis that classifies each referenced person or group by scale (individual, institution, or generalized class) and dimension-specific explanations that articulate the reasoning behind each score. At each step, the model is asked to consider alternative readings and flag its confidence as LOW when a reasonable coder could disagree. Three few-shot examples are included to calibrate the model's decision boundaries: a strongly populist post with high people, elite, and antagonism scores, a non-political post that should receive all zeros, and an ambiguous post where institutional criticism could plausibly be read as either targeted policy dissatisfaction or broader anti-elite attitude. This example structure is designed to discourage binary classification tendencies and encourage the model to use the full range of each scale.

The prompt is included as a systemprompt and appended with the to be annotated tweet. The model outputs its reasoning and scores in a json format.

4.4 Textanalysis (Discuss LLM, Prompt, Validation) [332w, 0.6p]

Computationally methods driven text analysis is based on linguistic concepts. A corpus of documents is divided in tokens as semantically meaningful units of analysis which are often words. The simplest text classification techniques are usually wordfrequency based dictionary methods. Some of which already mechanistically take context of wordappearence into account (Ribeiro et al. 2016). While simple machine learning classifiers like naive bayes models are statistically more elaborate, deeplearning models are able

to integrate interactions between tokens allowing for even better predictions. But deep learning architectures like convolutional neural networks trade variance for increased local bias through usage of filters (Sohil, Sohali, and Shabbir 2022). Transformer architectures implement modeling of long-range dependencies by implementing self attention on a micro level weighing the importance of every individual input token for other each individual input token (Cordonnier, Loukas, and Jaggi 2020). Qwen3-235B-A22B-Instruct-2507-FP8 is using more elaborate attention mechanisms like grouped query attention (Yang et al. 2025). The model also implements expert segmentation, is multilingual and has 8-bit floating weights (Yang et al. 2025).

4.5 Networkanalysis (Threads, Replynetwork, Egonetworks) [295w, 0.6p]

Construction of user network (line 287 in 10_create_network.R)

4.6 Summary [100w, 0.2 p]

- Hat armin den Datensatz schonmal irgendwo verwendet, sodass ich ihn zitieren könnte?

1. First write this data collection section, send to armin for refinement and clarification purposes.
2. write a refined proposal that focuses on politicians as egonets

==> look in data where politicians are in reply network. make sure to filter first ofr ONLY the trees coming from politicians

5 Results

5.1 Introduction [100w, 0.2p]

5.2 Textanalysis: Result [510w, 1p]

The text

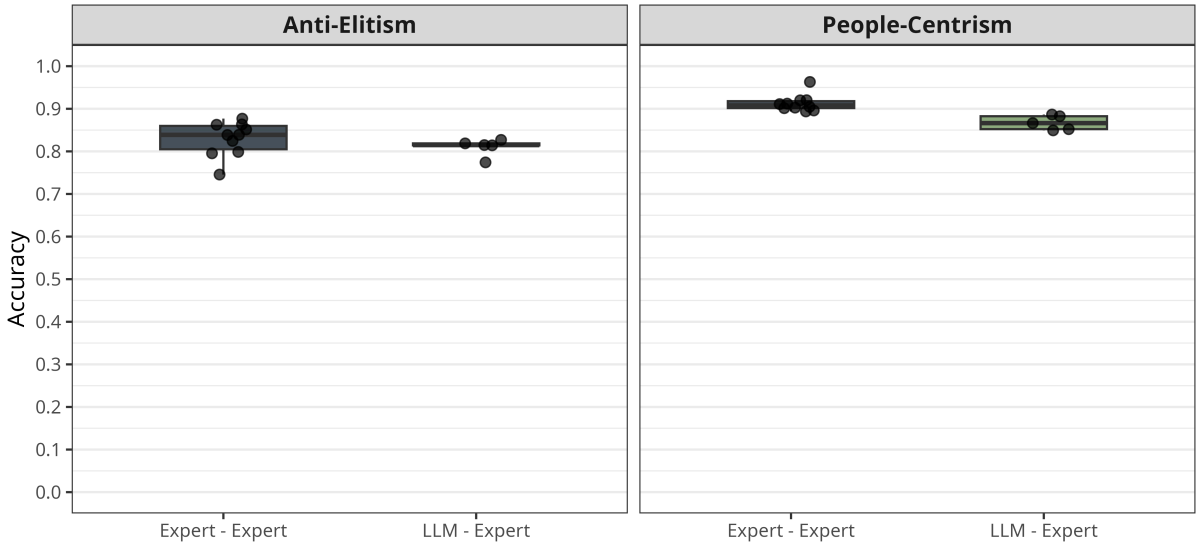

Figure 1: Prompt Performance - Pairwise accuracy for all expert-expert and LLM-expert rater combinations across anti-elitism and people-centrism.

Pairwise accuracy between binarized Qwen3-235B-A22B-Instruct-2507-FP8 ratings using the developed populism system prompt and five expert annotators from the PopBERT training corpus (Erhard et al. 2025). Each point represents one rater pair (expert-expert or LLM-expert). LLM ratings were collapsed from a 7-level Likert scale to binary scores (anti-elitism: score < 0; people-centrism: score > 0).

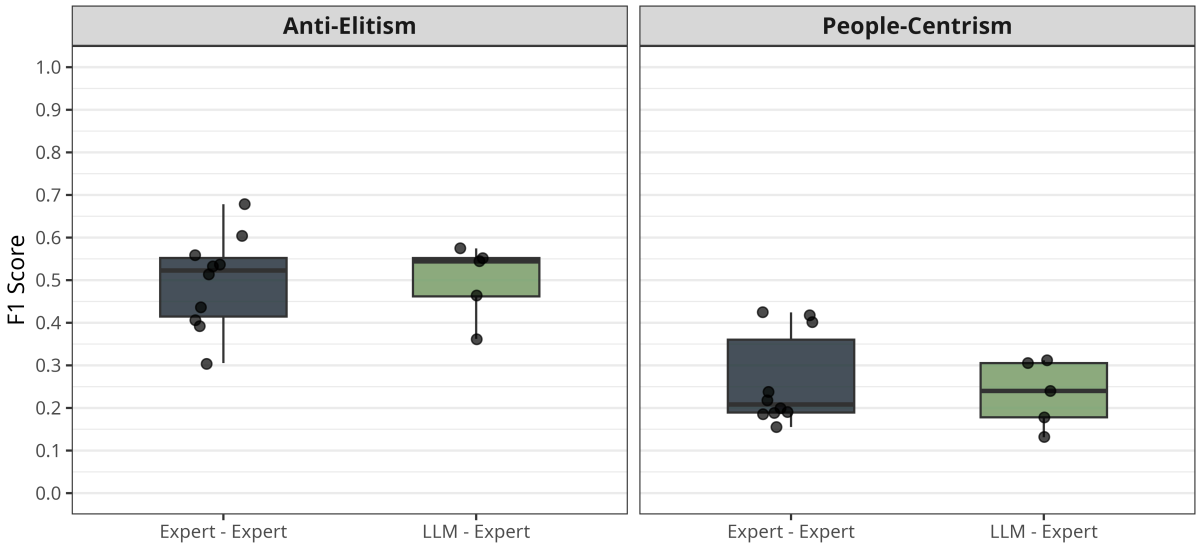

Figure 2: Prompt Performance - Pairwise F1 scores for all expert-expert and LLM-expert rater combinations across anti-elitism and people-centrism.

Pairwise F1 scores between binarized Qwen3-235B-A22B-Instruct-2507-FP8 ratings using the developed populism system prompt and five expert annotators from the PopBERT training corpus (Erhard et al. 2025). Each point represents one rater pair (expert-expert or LLM-expert). LLM ratings were collapsed from a 7-level Likert scale to binary scores (anti-elitism: score < 0; people-centrism: score > 0).

While the F1 score is not symetric for swapping labels it is symetric for swapping raters.

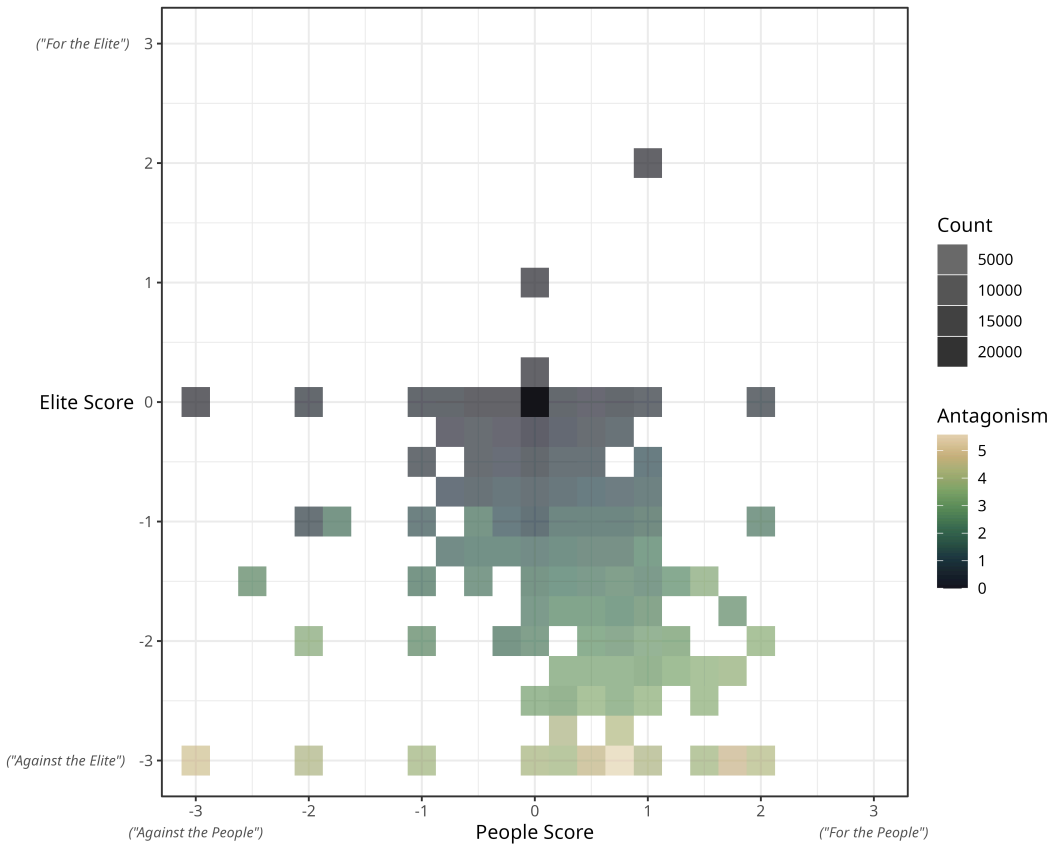

Figure 3: Title of the Figure

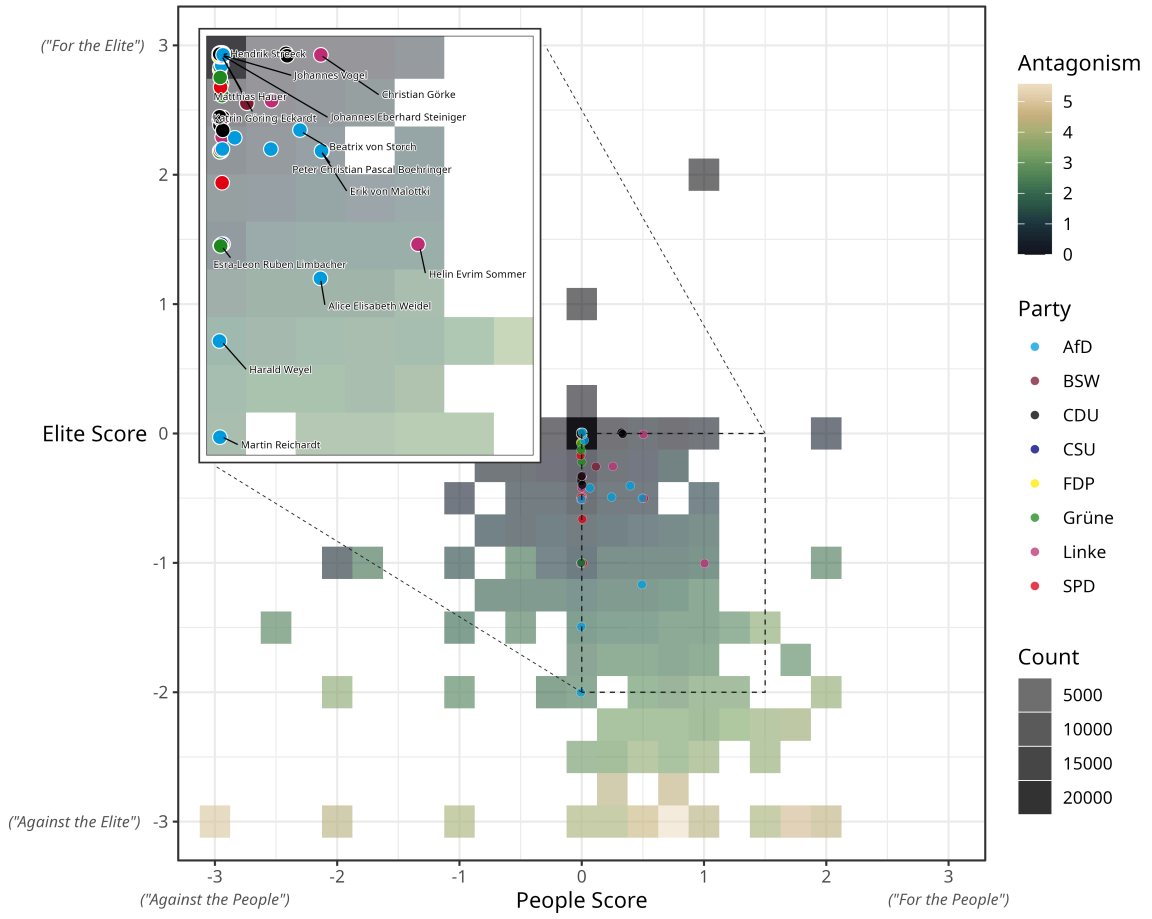

Figure 4: Title of the Figure

5.3 Network Analysis

5.4 Network: Descriptives & Structure [469w, 0.9p]

5.5 Network: Hierarchy Deepdive [386w, 0.7p]

5.6 Network: Egonetworks [457w, 0.9p]

5.7 H1: Model/Means [767w, 1.5p]

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6 Discussion

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6.6 Summary [100w, 0.2 p]

7 Conclusion

Appendix A: Additional Tables, Figures and Equations

Symetry of F1 Score for swapping raters

Given binary ratings $a_i, b_i \in \{0, 1\}$ from two raters, define:

$$\text{both_positive} = \sum_i a_i b_i$$

$$\text{only}_{b_{\text{positive}}} = \sum_i (1 - a_i) b_i$$

$$\text{only}_{a_{\text{positive}}} = \sum_i a_i (1 - b_i)$$

Treating A as ground truth:

$$\text{precision} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{b_{\text{positive}}}}, \quad \text{recall} = \frac{\text{both_positive}}{\text{both_positive} + \text{only}_{a_{\text{positive}}}}$$

Swapping roles swaps $\text{only}_{a_{\text{positive}}} \leftrightarrow \text{only}_{b_{\text{positive}}}$, which swaps precision and recall. Since F_1 is their harmonic mean:

$F_1 = \frac{2 \cdot \text{ precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \cdot \text{ recall} \cdot \text{precision}}{\text{recall} + \text{precision}}$

Multiplication and addition are commutative, so F_1 is invariant under the swap.

For swapping labels (0,1) F Scores are not symetric.

Appendix C: Prompt

Appendix B: Code

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